URANUS

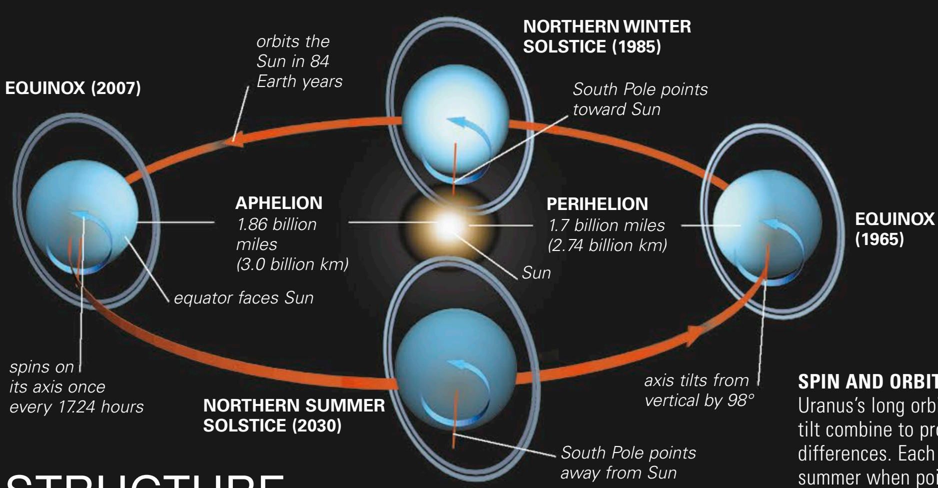
38–39 Gravity, motion, and orbits
68–69 Planetary motion
100–101 The history of the solar system
102–103 The family of the Sun

URANUS IS THE THIRD-LARGEST planet and lies twice as far from the Sun as its neighbor Saturn. It is pale blue and featureless, with a sparse ring system and an extensive family of moons. The planet is tipped on its side, so from Earth the moons and rings appear to encircle it from top to bottom. Uranus was the first planet to be discovered by telescope, but little was known about it until the Voyager 2 spacecraft flew past in January 1986.

PALE BLUE DISK
Voyager 2 images have been combined to show the southern hemisphere of Uranus as it would appear to a human onboard the spacecraft.

ORBIT

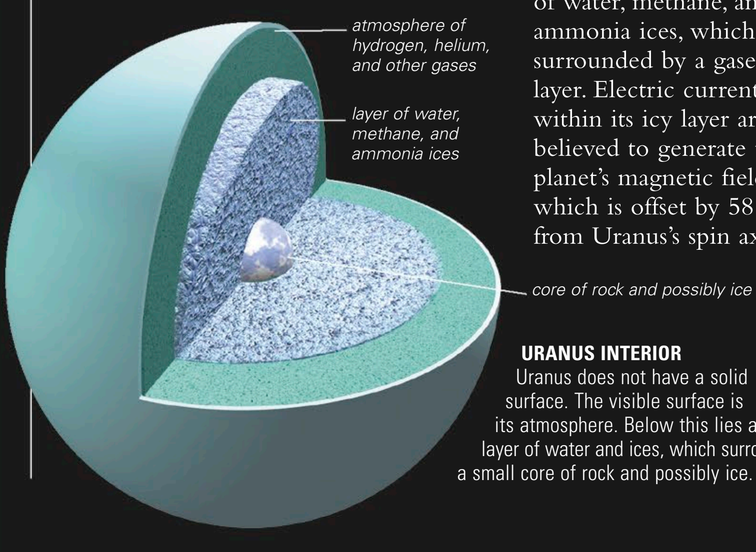
Uranus takes 84 Earth years to complete one orbit around the Sun. Its axis of rotation is tipped over by 98°, and the planet moves along the orbital path on its side. Uranus's spin is retrograde, spinning in the opposite direction of most planets. The planet would not have always been like this. Its sideways stance is probably the result of a collision with a planet-sized body when Uranus was young. Each of the poles points to the Sun for 21 years at a time, during the periods centered on the solstices. This means that while one pole experiences a long period of continuous sunlight, the other experiences a similar period of complete darkness. The strength of the sunlight received by the planet is 0.25 percent of that on Earth. When Voyager encountered Uranus in 1986, its south pole was pointing almost directly at the Sun. Uranus's equator then became increasingly edge-on to the Sun. After 2007, it has progressively turned away, and the north pole will face the Sun in 2030.



SPIN AND ORBIT
Uranus's long orbit and its extreme tilt combine to produce long seasonal differences. Each pole experiences summer when pointing toward the Sun and winter when it is pointing away. At such times, the pole is in the middle of Uranus's disk when viewed from Earth. At the equinoxes, the equator and rings are edge-on to the Sun.

STRUCTURE

Uranus is big. It is four times the size of Earth and could contain 63 Earths inside it, yet it has only 14.5 times the mass of Earth. So the material it is made of must be less dense than that of Earth. Uranus is too massive for its main ingredient to be hydrogen, which is the main constituent of the bigger planets, Saturn and Jupiter. It is made mainly of water, methane, and ammonia ices, which are surrounded by a gaseous layer. Electric currents within its icy layer are believed to generate the planet's magnetic field, which is offset by 58.6° from Uranus's spin axis.



URANUS INTERIOR
Uranus does not have a solid surface. The visible surface is its atmosphere. Below this lies a layer of water and ices, which surrounds a small core of rock and possibly ice.

URANUS PROFILE

AVERAGE DISTANCE FROM THE SUN 1.78 billion miles (2.87 billion km)	**ROTATION PERIOD** 17.24 hours
CLOUDTOP TEMPERATURE –364°F (–220°C)	**ORBITAL PERIOD (LENGTH OF YEAR)** 84 Earth years
DIAMETER 31,763 miles (51,118 km)	**MASS (EARTH = 1)** 14.5
VOLUME (EARTH = 1) 63.1	**GRAVITY AT CLOUDTOPS (EARTH = 1)** 0.89
NUMBER OF MOONS 27	**SIZE COMPARISON**
OBSERVATION Uranus's remote location makes it a difficult object to view from Earth. At magnitude 5.5, it is just visible to the naked eye and looks like a star. There is no perceptible change in brightness when Uranus is at opposition.